HACKSPHERE 3.0 · TEAM **QUANTIX**

GridPilot

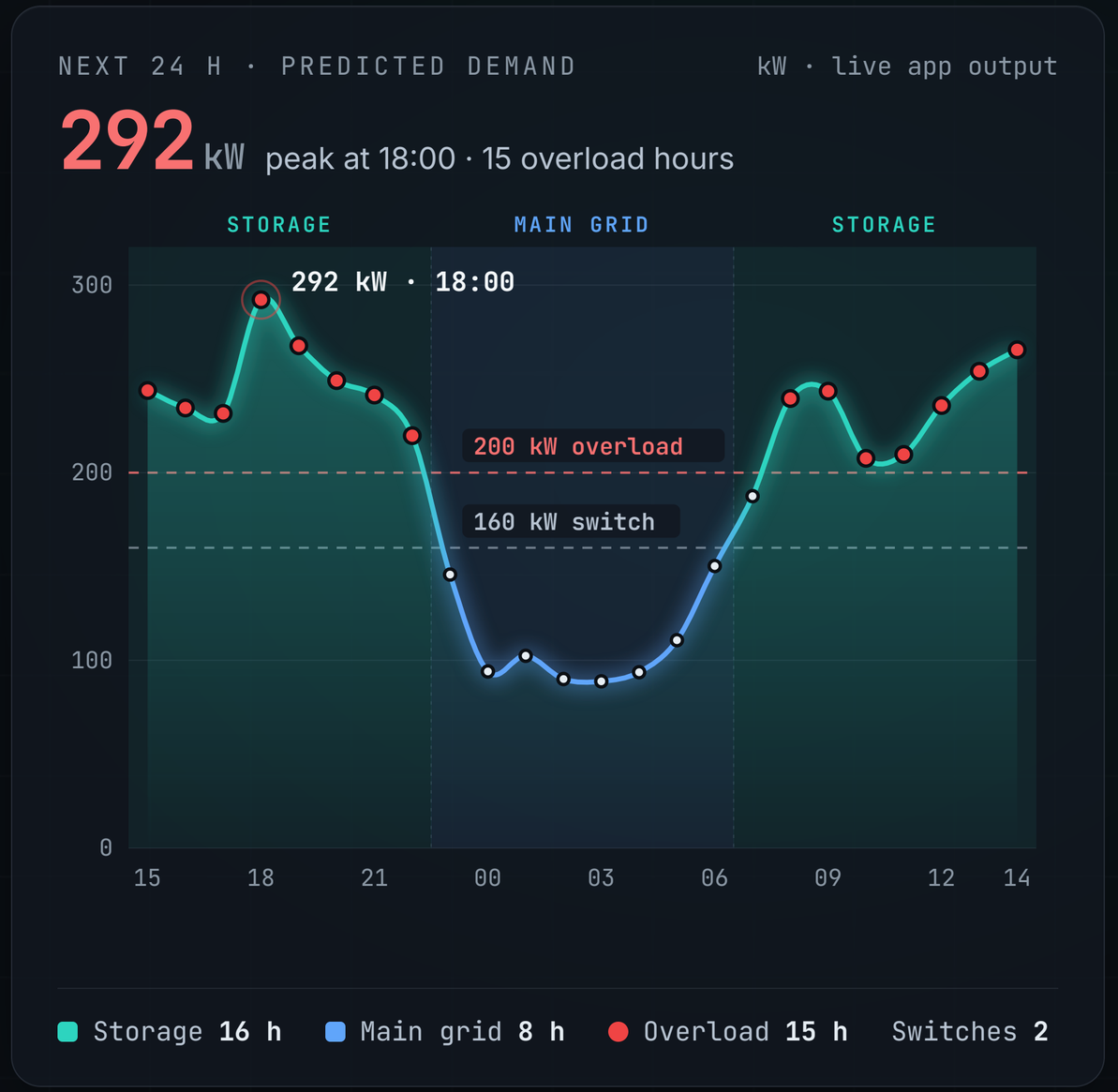
Forecast tomorrow's load.
Switch before it spikes.

Built on **Zoho Catalyst** · AppSail + QuickML

LIVE APP

Scan to open the dashboard

gridpilot-50047251149.development.catalystappsail.in



THE PROBLEM · ONE CALL PER HOUR

Every hour, a microgrid must choose: main grid or battery.

EVERY HOUR

24 decisions / day

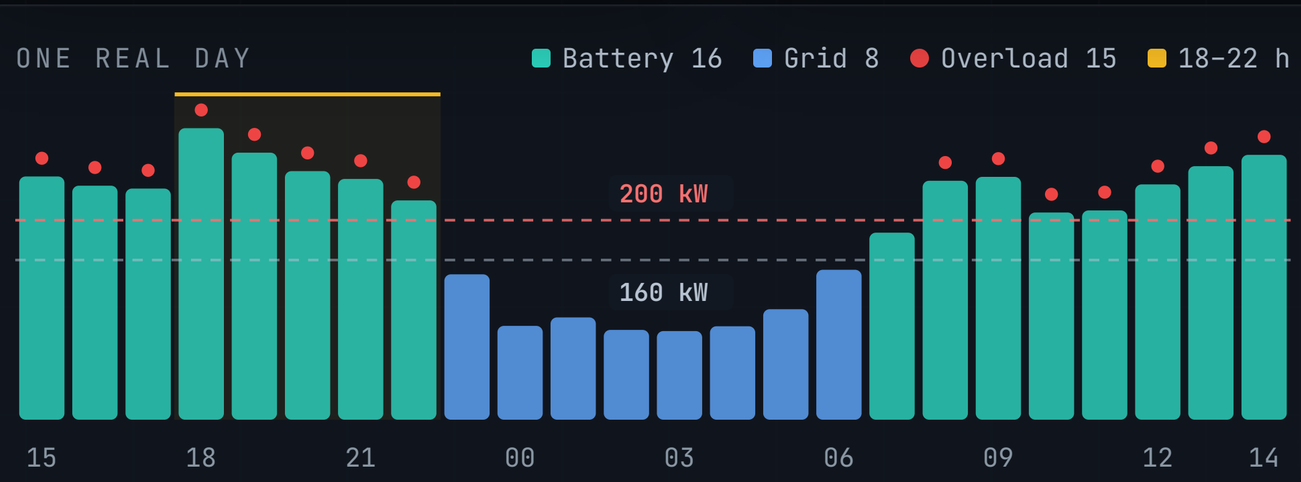
EVENING PEAK

18–22 h window

OVERLOAD RISK

≥200 kW

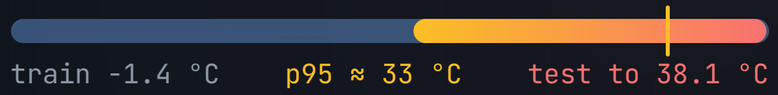
ONE REAL DAY



WHY NAIVE MODELS FAIL

236 / 1,000

test hours are hotter than 95% of the training data. The model must extrapolate.



One pipeline. Weather in, signals out.

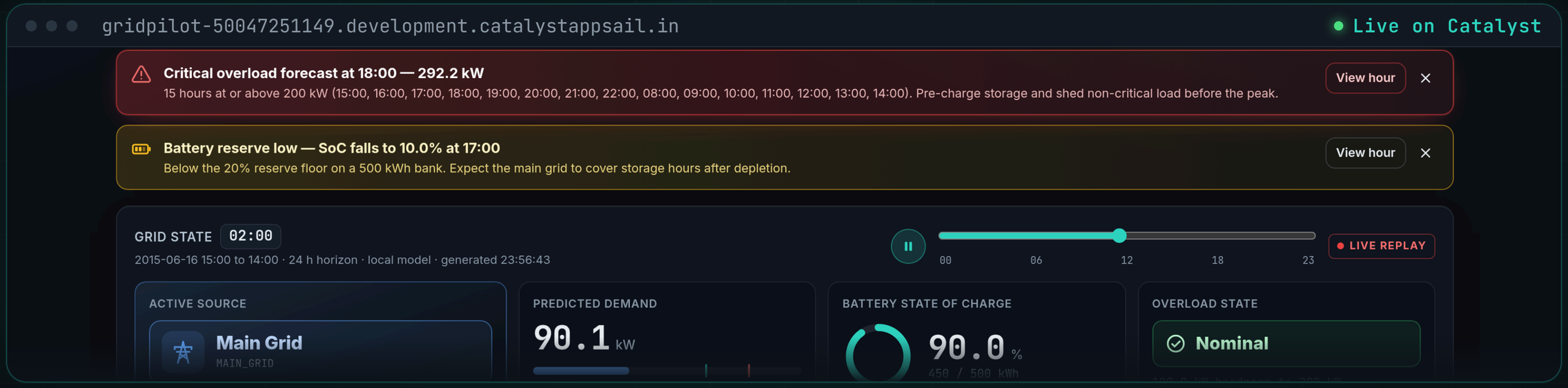
01
Ingest
Form or CSV, hourly weather

02
Clean
Validate, impute, clip

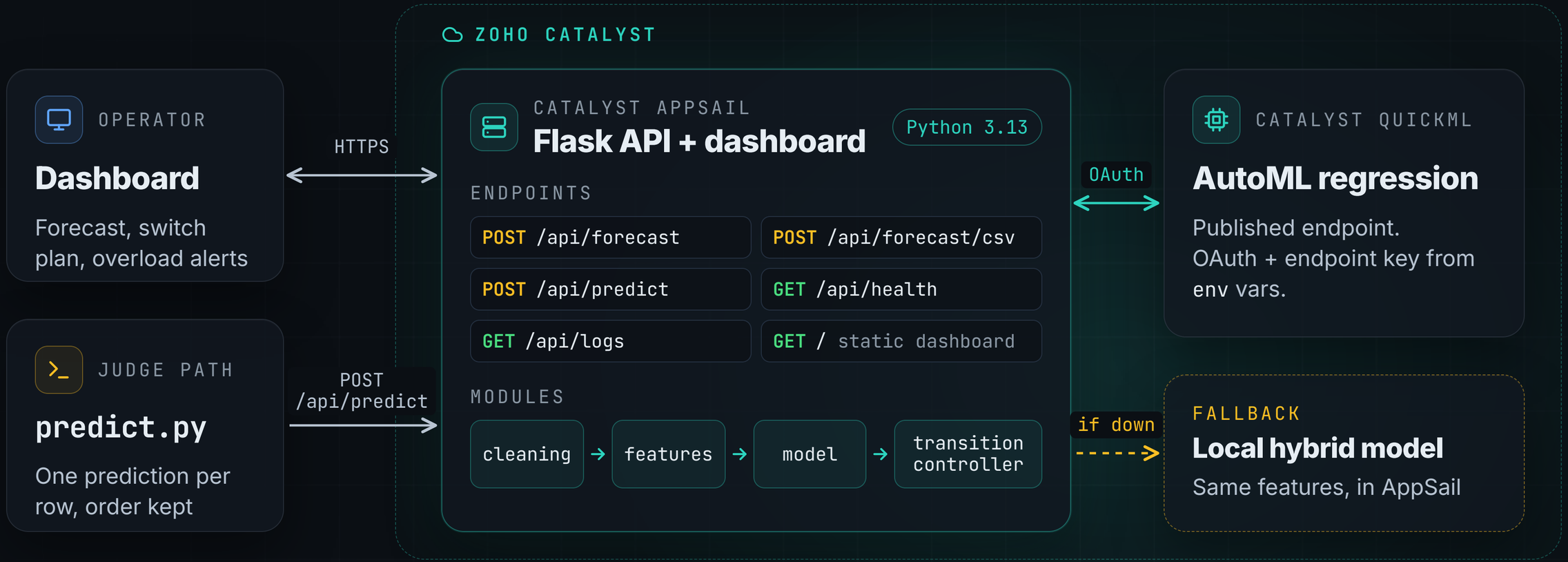
03
Forecast
QuickML + hybrid model, 24 h

04
Decide
Transition controller

05
Act
Dashboard, alerts, dispatch signals



Three parts. All on Catalyst.



Secrets in env vars, never in git

Deployed with Catalyst CLI

4,000 hours in.

Cleaned the same way at **train** and **predict** time.

SCHEMA · HOURLY

- timestamp
- ambient_temp_c
- humidity_pct
- solar_irradiance_wm2

↓

- target_demand_kw

4,000 rows · 2015

MESSY UPLOAD → CLEANED

raw value shown after ←

timestamp	temp °C	humid %	W/m²	fix
2015-06-17 15:00:00	35.7	56.4	169	ok
2015-06-17 16:00:00	35.0 ← blank	54.7	112	imputed
2015-06-17 17:00:00	34.3	100 ← 140	52	clipped
2015-06-17 18:00:00 ← 17-06-2015 18:00	35.4	56.9	0 ← -5	normalised clipped
~~not a date~~	~~32.1~~	~~56.4~~	~~2~~	dropped
2015-06-17 20:00:00	30.0	59.1	0	ok

LIVE CSV TEST

POST /api/forecast/csv

6 → 5

rows in rows out

1 dropped

1 imputed

2 clipped

FEATURES hour-of-day · weekday/weekend · sin · cos hour · cooling T-22 °C · heating 15 °C-T · humidity · solar · peak flag 18-22 h

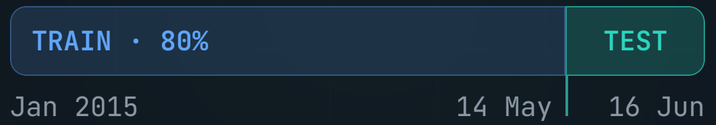
Tested on weeks it never saw.

HYBRID · RMSE

9.70 kW

MAE **7.74 kW**

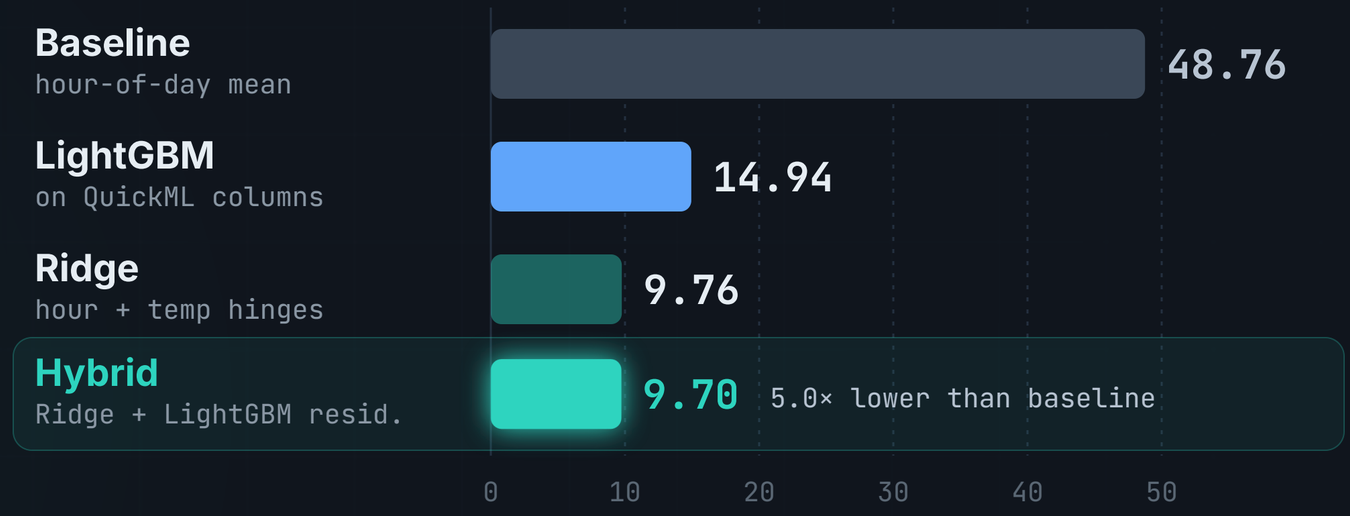
CHRONOLOGICAL HOLD-OUT



No shuffling: trained on Jan–May, tested on the hotter weeks after.

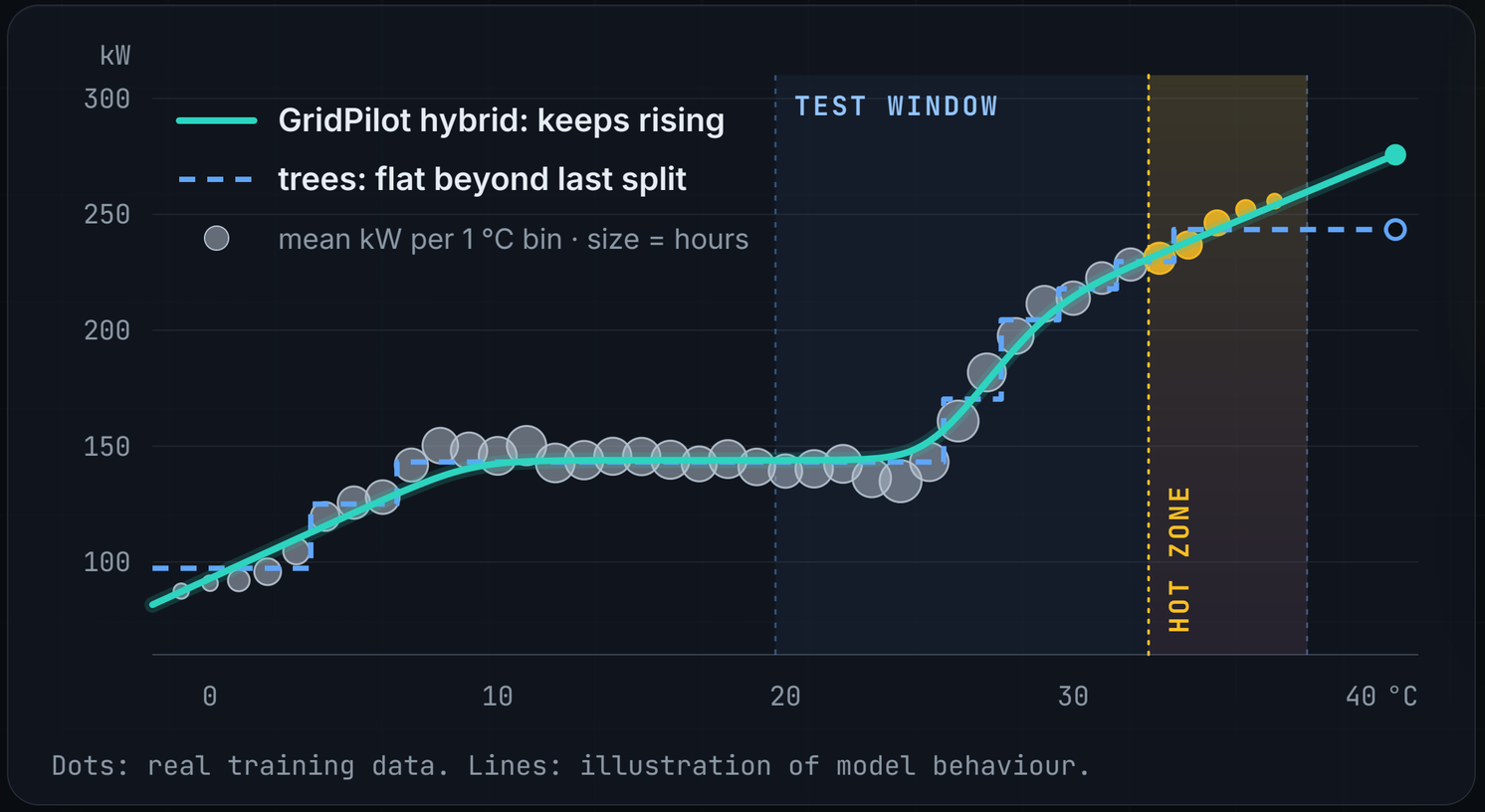
RMSE ON HOLD-OUT · KW

↓ lower is better



Why hybrid: Ridge base carries temperature beyond the training range; gradient-boosted trees fix what's left.

The test weeks run hotter than 95% of what the model trained on.



Dots: real training data. Lines: illustration of model behaviour.

HOT ZONE

236 / 1,000

test hours sit in the hot zone

TRAINING P95	32.6 °C
TEST WINDOW	19.7–38.1 °C

Trees cap at their last split. A temperature hinge **keeps climbing** with the heat.

Trained and published in Catalyst QuickML.

01 · DATASET
gridpilot_quickml_train
4,000 rows · 14 cols · quality 10/10

02 · CLEAN
Mean Imputer
fills any gaps per column

03 · SCALE
Min-Max Normalisation
every feature to 0-1

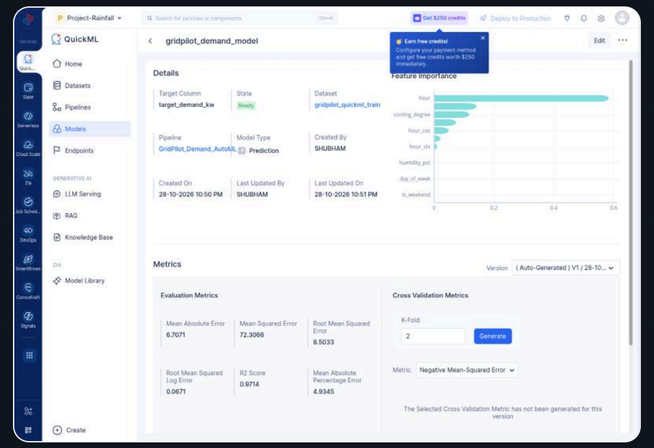
04 · MODEL
GB Regression
gradient-boosted trees

05 · SERVE
Published endpoint
gridpilot_demand_endpoint

R^2 SCORE	RMSE	MAE	MAPE
0.971	8.50 kW	6.71 kW	4.93 %

TOP FEATURES
QuickML importance, ranked

1. hour
2. cooling_degree
3. hour_cos



QuickML model page

POST /quickml/v1/.../endpoints/predict · **HTTP 200** · **197.89 kW**

Three rules. Every edge tested.

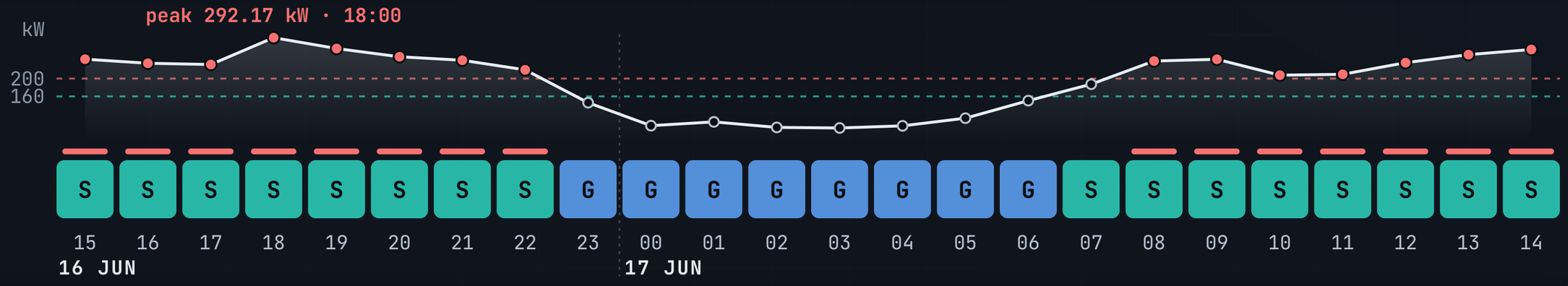
RENEWABLE STORAGE
demand > **160 kW**
OR **18:00–22:00** inclusive

MAIN GRID
otherwise
≤ 160 kW and outside the peak

CRITICAL OVERLOAD ALERT
demand ≥ **200 kW**
independent of mode

REAL 24-H FORECAST · **16 JUN 15:00 → 17 JUN 14:00**

S Storage 16 h G Grid 8 h Overload 15 h 2 transitions



160.00 kW → Main Grid	160.01 kW → Storage	200.00 kW → Alert	22:00 → Storage	23:00 → Main Grid	9 boundary tests passing

See the next 24 hours at a glance.



292 kW	16 h 8 h	15 h	2
peak @ 18:00	storage / grid	overload hours	switches

1. **Critical overload banner**
   Hour and kW, flagged before the peak
2. **Active source + battery SoC**
   What is live now, charge left
3. **24-h dispatch timeline**
   Storage, grid, overload per hour
4. **KPI cards**
   Peak, energy, switches, min SoC
5. **CSV / manual ingestion**
   Upload a file or type the hours

Built for the scorecard.

35% Model accuracy

9.70 kW

RMSE · hybrid model

- ✓ QuickML R^2 0.971
- ✓ 5× better than baseline

35% Deployment & functionality

Live

on Zoho Catalyst AppSail

- ✓ CSV + manual input
- ✓ Public REST API

20% Transition controller

3 rules

storage · grid · overload

- ✓ 9 boundary tests pass
- ✓ 160 kW · 18–22 h

10% UI & UX

24 h

dispatch timeline

- ✓ Responsive dashboard
- ✓ Overload + battery alerts

hacksphere-gridpilot · fresh clone

```
$ python test_eval.py
SUCCESS
$ python predict.py --input sample_test_input.csv
Wrote 1000 predictions  # 500-row chunks · 30 s timeout · 1 retry
```

TEST SUITE

24 passing

15 API · 9 controller boundary

QUANTIX.

YC **Yashvi Chotalia** TEAM LEADER

SS Shubham Sahu AJ Aditya Jadhav SH Samhita Hejmadi

Scan to open the live app

Thank you.

LIVE APP gridpilot-50047251149.development.catalystappsail.in

REPO github.com/https-shubhamsahu/hacksphere-gridpilot

LEADER github.com/YashviChotalia

WHAT'S NEXT · ROADMAP

01 Live meter feeds via Catalyst Event Listeners

02 Forecast history in Catalyst Data Store

03 Per-site models through QuickML pipelines

04 Operator SMS / Signals alerts before the overload hour